Probe-guided connection rebalancing: greedy or DQL control

Active sockets: events / ACKs / stats. Dashed probe: telemetry only until guarded promotion.

Smart client

Connections + policy

Telemetry (10 s)
Counts + backlog
Pod stats + ACKs

Policy selector
Greedy or DQL

$o_u(t)$ → a^{rebN}, a^{rebG}, a^{mig}

Bounded action
Rules or Q-values

DQL pictogram when enabled

Guarded executor
A: Reassociate k
B: Promote same c_p

Source socket c_h
High $N_{u,c}$ or $G_{u,c}$

A: a^{rebN} / a^{rebG}

Existing socket c_l
Lower-load destination

1 Retained probe c_p
Authenticated; stats only
Selection + guards first

Kubernetes

Service / LB

c_h

c_l

2 LB assigns
probe pinned to pod

c_p

Subscriber service cluster

Persistent delivery endpoints

Active sockets / queue access
Probe / sampled telemetry
Selected action / guarded control

Source pod i_h
Current delivery owner
Reports X_{i_h}, B_{i_h}, H_{i_h}

Existing lower-load pod i_l
A: receives k through c_l
Both active sockets remain open

3 Candidate pod i^{probe}_u
LB endpoint of probe c_p
No delivery before promotion

Durable queues

Per subscription
Fetch / events: owners only

$Q_{u,k1}$ $b_{u,k1}$

$Q_{u,k}$ Fixed

$Q_{u,k2}$ $b_{u,k2}$

Same queue IDs
No payload copy

Queues do not move

A Subscription rebalancing

a^{rebN} / a^{rebG}

A1 Select action Resolve k; guards → **A2 Prepare k** Delivery disabled → **A3 Quiesce k** ACK barrier → **A4 Activate k** Target receipt → **A5 Detach source k** Both sockets stay open

B Probe migration

a^{mig}

Conditional promotion

B1 Policy Select a^{mig} → **B2 Guards** Admissible? → (yes) **B3 Promote** First binding → **B4 Standby** All receipts → **B5 Quiesce** Stop source → **B6 Drain** Fresh zero → **B7 Activate** Receipts → **B8 Detach** Retire c_h

No: keep source; retain / recycle c_p

Connection: active source + probe → guarded handoff → same probe active; source retired

Subscriptions: same IDs + same queues → standby → quiesce / drain → activate / detach

Handoff checks: standby receipts; source quiescence; fresh zero in-flight; target activation receipts